5. (*a*) Explain step angle or microstepping of a stepper motor. 2

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(*b*) Derive an expression for linear induction motor in terms of loadings and linear dimensions. 7

(*c*) Briefly explain the following characteristics of stepper motors : 7

(*i*) Torque displacement characteristics

(*ii*) Torque stepping rate characteristics

(*d*) What is reluctance motor? Show that it can develop unidirectional torque at only one speed. 7

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324652(24)

BE (6th Semester)

Examination, April - May, 2016

(New Scheme)

Electrical Machines - III

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Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 28

Note : Attempt part (*a*) and two parts (*b*), (*c*) and (*d*) of each questions. Part (*a*) carries two marks and (*b*), (*c*) and (*d*) carry seven marks each.

1. (*a*) What are the assumptions of an Ideal Synchronous Machine? 2

(*b*) Derive transient power angle equation of a Synchronous Machine. 7

(*c*) For ideal synchronous machine, show that torque is given by

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$$T = \frac{\text{Poles}}{2}(\lambda_d i_q - \lambda_q i_d) \times \frac{\text{Phases}}{2}$$ 7

(*d*) Show that self and mutual inductances of stator phases of three-phase two pole synchronous machine can be put in the form of

$$\alpha_{aa} = L_{al} + Lg_0 + Lg_2 \cos 2\theta$$

$$\alpha_{bc} = -0{\cdot}5\, Lg_0 + Lg_2 \cos 2\theta$$ 7

2. (*a*) Enlist the main causes of single phasing of 3-phase induction motor. 2

(*b*) From *d* and *q* axis voltage expression, derive the equivalent circuit of three-phase induction motor for balance input. 7

(*c*) Park transformation for three-phase current is given by 7

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$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 1 \\ \cos(\theta-120) & -\sin(\theta-120) & 1 \\ \cos(\theta+120) & -\sin(\theta+120) & 1 \end{bmatrix} \begin{bmatrix} I_d \\ I_q \\ I_o \end{bmatrix}$$

Test the above transformation for power invariancy.

(*d*) Using transient analysis, discuss the initial current of induction motor on occurence of three-phase fault. 7

3. (*a*) Why is centrifugal switch provided in single-phase induction motor? 2

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(*b*) The following data relates to test on a 110 V, 150 watts, 50 Hz, 6-pole single-phase induction motor :

No load test	110 V	2·7 A	63 watts
Blocked rotor test	55 V	5·8 A	212 watts

The stator winding resistance is 2·5 ohms and during the blocked rotor test the starting winding is open. Determine the equivalent circuit parameters. Also find the core, friction and windage losses. 7

(*c*) A 230 V, 4-pole, 50 Hz single-phase induction motor has the following constants and losses :

$r_1 = 2{\cdot}3\ \Omega$, $r_2 = 4{\cdot}2\ \Omega$, $x_1 = 3{\cdot}2\ \Omega$, $x_2 = 3{\cdot}2\ \Omega$

$x_m = 74\ \Omega$, core loss = 98 watts.

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Friction and windage loss = 30 watts

If this motor is running with a slip 0·05 at rated voltage and frequency, then compute the stator current, p.f., power output, torque and efficiency with its auxiliary winding open. 7

(*d*) Develop and draw the torque-slip characteristics of single-phase induction motor based on double field revolving theory. 7

4. (*a*) Write down the application of universal motor. 2

(*b*) Describe the methods of improving commutation. 7

(*c*) Discuss the action of commutator as a frequency converter. 7

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(*d*) Derive an expression for–

(*i*) Transformer voltage;

(*ii*) Speed voltage induced in armature of a.c. commutator machine in case the field flux is sinusoidal. 7